

CHAPTER 1

INTRODUCTION TO THE MARINE ENVIRONMENT

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

We live on a water planet, with about 71% of the Earth's surface being covered by salt water to an average depth of 3.8 km. This introductory chapter, which presents an overview of the complex discipline of marine ecology, is divided into six parts. The first part addresses the unusual qualities of water and how those properties are modified in seawater (PROPERTIES OF WATER). The second part covers the general concepts of BASIC OCEANOGRAPHY, and the third reviews the fundamental principles of ecology (SOME ECOLOGICAL PRINCIPLES). The fourth part then explores the unique aspects of LARVAE AND LARVAL ECOLOGY. The penultimate part examines the differences between terrestrial and marine ecosystems (COMPARISON OF TERRESTRIAL AND MARINE ECOSYSTEMS), while the final part presents an important nuts-and-bolts view of the various DIVISIONS OF THE MARINE ENVIRONMENT.

I. Properties of Water

The first part of this chapter is separated into three sections:

- Chemical Composition
- Physical and Chemical Properties
- Seawater

4°C	cohesion	oxygen minimum zone
acidity	conductivity	pH scale
adhesion	density	polar molecule
alkalinity	freezing-point depression	practical salinity units (psu)
bicarbonate ion	heat capacity	proton, H ⁺
buffer	heat of vaporization	refractive index
carbonate equilibrium system	hydrogen bonds	salinity
carbonate ion	hydrogen ion	surface tension
carbonic acid	hydroxyl ion	universal solvent
chlorinity	latent heat of fusion	viscosity

- Students should understand the chemistry of the water molecule and appreciate how hydrogen bonds form among adjacent water molecules. The importance of these hydrogen bonds can be illustrated by comparing the freezing and boiling points of water, ammonia, and methane, three molecules with similar molecular weights.

The following table compares the freezing and boiling points (°C) of H₂O, NH₃, and CH₄:

Molecule	MW	Freezing Point	Boiling Point
Water	18	0.0	100.0
Ammonia	17	−77.7	−33.4
Methane	16	−182.5	−164.0

- While rather mundane, the properties of cohesion (surface tension and viscosity), high heat of vaporization, high latent heat of fusion, and high heat capacity must be understood and their importance appreciated by students. Some simple demonstrations might serve to illustrate these principles.

2 Instructor's Manual for Marine Biology

Cohesion may be demonstrated by floating a fine needle (or wire) on the surface of water in a container and then disrupting the equilibrium (surface tension) by adding a drop of dish soap. Some practice is recommended before you try this demonstration before the class! As a backup, use a videotape of a demonstration that has worked. To initiate a discussion of the high heat of vaporization and latent heat of fusion, you can pose several questions concerning how perspiration cools a person (heat of vaporization) and why fruit growers in southern states spray water on their crops during the nights when a cold snap is expected (latent heat of fusion). The high heat capacity of water may be illustrated by plotting the temperature of air and water held in vessels of the same volume after they have been placed in a refrigerator.

3. The density relationships of pure water and seawater at temperatures of less than 5°C are interesting. Pure water reaches its maximum density at 4°C, but the density of seawater continues to increase until it freezes at about -1.9°C . However, in that process, the salts are excluded from the ice matrix that is forming, and the ice that results is composed of pure water. The salts that are left behind sink into the depths as a slow current of cold, dense, saltier water.
4. The meaning of salinity is often misinterpreted by a few students. A simple demonstration in which 35 grams of table salt is set out next to 965 grams of distilled water may help them understand the relationship. This, of course, produces a salinity of 35‰. (N.B.: ‰ = practical salinity units, or psu.) Additional details on the fact that most ($\approx 99\%$) of the dissolved material in seawater is composed of only six inorganic ions (chloride, sulfate, sodium, potassium, magnesium, and calcium) should be noted. Of the remaining less than 1%, about 75% is composed of bicarbonate, bromide, boric acid, and strontium. Much of the minute fraction that remains contains phosphate, nitrate, and silicon dioxide.
5. The solubility of oxygen in water is not great, and two factors act to further decrease it: temperature and salinity. At 0°C , pure water can hold about 14 mg/L of oxygen, while seawater can only hold only 8 mg/L. For both conditions, the water is 100% saturated with oxygen. Photosynthesis can increase the amount of oxygen that water can hold; thus, the water can become supersaturated. Respiration, of course, will lower the concentration; at times, the oxygen concentration can approach 0 mg/L. Organisms living in anoxic water must use alternative metabolisms to aerobic respiration; that is, hydrogen/electron acceptors other than molecular oxygen are used. (See the following point.)
6. In fermentative metabolisms, pyruvate (pyruvic acid), produced by glycolysis, is the final hydrogen/electron ($\text{H}^{+}/\text{e}^{-}$) acceptor. End products of such pathways include a variety of organic alcohols and acids (for example, butanol, ethyl alcohol, acetic acid, propionic acid, and lactic acid) and often CO_2 . Fermentation does not involve the Krebs cycle (tricarboxylic acid cycle).

Another alternative metabolism is anaerobic respiration. Anaerobic respirations resemble aerobic respiration in that they possess electron transport chains, but this process is less efficient, and, of course, molecular oxygen is not used as a final $\text{H}^{+}/\text{e}^{-}$ acceptor. The alternative $\text{H}^{+}/\text{e}^{-}$ acceptors that may be used depend on the organism in question (bacteria), but they include sulfur, nitrogen compounds, and CO_2 . Their end products are H_2S , NH_4^{+} , and CH_4 , respectively.

The term anaerobic metabolism (or respiration) is often used in a casual way to refer to what would otherwise be called fermentation ($\text{H}^{+}/\text{e}^{-}$ addition to pyruvate, making lactic acid, etc.). This biochemical reaction occurs in the legs of runners when oxygen is in short supply and the muscles go into oxygen debt. However, this biochemical reaction is the same as lactic acid production in yogurt fermentation. Perhaps a better way to distinguish true anaerobic respiration from oxygen debt is to use the form anaerobic glycolysis for the former.

"That which we call a rose by any other name would smell as sweet."

—Shakespeare, *Romeo & Juliet*, 2:2:43–44.

7. The solubility of carbon dioxide is different from that of oxygen. The carbonate equilibrium system is an important concept to master, but it is not always intuitively obvious what happens during shifts in equilibrium. It is correct to say that if CO_2 is removed (degassed) from water, then the concentrations of bicarbonate and carbonate will shift, with a concomitant increase in the production of CO_2 as H_2CO_3 . However, during times of intense photosynthesis in which bicarbonate is used as a source of CO_2 , the equilibrium will shift to CaCO_3 (limestone or marl), which will precipitate out.

In warm shallow seas and hard-water lakes, the precipitation of limestone may be seen in aerial views. This process is called whiting.

II. Basic Oceanography

The basic concepts of oceanography and geology, as they inform our understanding of the shape of the oceanic basins, are presented in this part of Chapter 1, which is divided into four sections:

- Geography and Geomorphology of the Oceans
- Plate Tectonics
- Temperature and Vertical Stratification
- Water Masses and Circulation

abyssal plain	fetch	subduction
Arctic Ocean	gyres	submarine ridges
Atlantic Ocean	homeotherm	temperature stratification
biogeographical zones	hydrothermal vents	thermocline
Challenger Deep	Indian Ocean	thermohaline circulation
coastal upwelling	internal waves	tides
cold temperate zone	La Niña	transform faults
continental crust	mantle	trenches
continental drift	Marianas trench	tropical zone
continental shelf	metabolism	trough
continental slope	northeast trades	tsunamis
Coriolis effect	Great Oceanic Conveyor Belt	upper-water masses
deepwater masses	oceanic crust	volcanic island arcs
deep-focus earthquakes	oceanography	warm temperate zone
easterlies	Pacific Ocean	wave height
ectotherm	plates	wave period
Ekman spiral	poikilotherm	wavelength
El Niño	polar zone	waves
endotherm	pycnocline	Weddell Sea
ENSO	seafloor spreading	westerlies
equatorial	seamounts	wind belts
equatorial upwelling	southeast trades	

1. The first section of this part reviews, in general terms, the morphometry of oceans, from contact with the continents to the abyssal plains. A 2 × 2 color slide set (*Oceans and Shorelines*) is available from Ward's Natural Science Establishment, Inc.
2. Instructors may wish to add more material than is covered in the text on the theory of continental drift and the unifying theory of plate tectonics to their introduction to the world's oceans. While this topic is somewhat outside the general discipline of marine biology and is covered in introductory geology courses, it is appropriate to pay some attention to it here.

Coverage of this topic might start with a historical perspective of the German meteorologist Alfred Wegener and others who argued for continental drift. A brief review of the evidence that supports the theory would include (1) the general fit of continental coastlines; (2) the matching of fossil and stratigraphic markers across adjacent continents; (3) the fit of paleoclimatic data, especially of glaciation; and (4) the matching of paleomagnetic indicators. Finally, an abbreviated formulation of the theory and its predictions, as well as its problem areas, could be presented.

Any good introductory geology text will provide an overview of this topic, but see also the forthcoming list of citations. A variety of media on the concepts of PLATE TECTONICS is available from Wards Natural Science Establishment, Inc., among other supply houses. Although a tad old, a 60-minute video entitled *Secrets of Underwater Volcanoes* (1994) is available from Public Media Education (ISBN 0-7800-1601-7). The text by McGeary and Plummer (1998) comes with a CD-ROM on plate tectonics, and there is also a video series (*Earth Revealed*) available from Annenberg that

4 Instructor's Manual for Marine Biology

covers this and many other topics that inform aspects of marine biology. The text and the video series more or less coordinate with each other. The following is a list of additional sources:

- Allègre, C.J., and S.H. Schneider. 1994. The evolution of the earth. *Sci. Am.* 271(4):66–75.
- Anon. (The 1996 Lo'ihi Science Team). 1997. Researchers rapidly respond to submarine activity at Lo'ihi volcano, Hawaii: EOS. *Trans. Am. Geo. Union* 78(22):229–233.
- Bonatti, E. 1994. The Earth's mantle below the oceans. *Sci. Am.* 270(3):44–51.
- Courtillot, V., and G.E. Vink. 1983. How continents break up. *Sci. Am.* 249(1):43–49.
- Davidson, K., and A.R. Williams. 1996. Under our skin. *Nat. Geo.* 189(1):100–111.
- Dvorak, J.J., C. Johnson, and R.I. Tilling. 1992. Dynamics of Kilauea Volcano. *Sci. Am.* 267(2):46–53.
- Francheteau, J. 1983. The oceanic crust. *Sci. Am.* 249(3):114–129.
- Hékinian, R. 1984. Undersea volcanoes. *Sci. Am.* 251(1):46–55.
- Hyndman, R.D. 1995. Giant earthquakes of the Pacific northwest. *Sci. Am.* 273(6):68–75.
- Kious, W.J., and R.I. Tilling. 1996. *This dynamic earth*. USGS Information Services, Box 25286, Building 810, Denver Federal Center, Denver, CO 80225. <http://pubs.usgs.gov/publications/text/dynamic.html>.
- Lonsdale, P., and C. Small. 1991. Ridges and rises: a global view. *Oceanus* 34(4):26–35.
- Lutz, R.A., and R.M. Haymon. 1994. Rebirth of a deep-sea vent. *Nat. Geo.* 186(5):114–126.
- McGeary, D., and C.C. Plummer. 1998. *Physical geology: earth revealed*. 3d ed. Boston: WCB McGraw-Hill.
- Murphy, J., and R. Nance. 1992. Mountain belts and the supercontinental cycle. *Sci. Am.* 266(4):84–91.
- Oceanus*. (Reports on research from the Woods Hole Oceanographic Institution). 1998. Vol. 41 (1 & 2): The Mid-Ocean Ridge, parts 1 & 2.
- Parks, N. 1994. Exploring Loihi: the next Hawaiian island. *Earth* 3(5):56–63.
- Scientific American*. 1993. The dynamic earth. *Sci. Am.* 249(3):the entire issue.
- Simpson, S. 2000. Killer waves on the East coast? *Sci. Am.* 283(4):16–17.
- Tairo, A. 1993/94. When plates collide. *Oceanus* 35(4):95–98.

3. Development of the thermocline is an important concept that may be explored in a demonstration or laboratory setting. This can be done using a 10-gallon aquarium filled to within 2–3 cm of the top with cold water (ca. 4°C). Mark off the depth of the tank along one side, and set 5 to 10 thermometers at appropriate depths. An IR lamp held 15 cm above the water by a ring stand will serve as an artificial sun, while a hair dryer (or blower), set on the cool-air setting, provides the wind energy sufficient to distribute the heat into the water column. Be careful to prevent water from splashing on the lamp, and do not allow the lamp to melt the plastic edging of the aquarium. The artificial wind and sunlight can be alternated or run simultaneously; these conditions will produce thermoclines with different shapes.

Once the thermocline begins to develop, turn the blower off, gently add (sprinkle) a small amount of a powdered dye such as methylene blue on the surface of the water. Distribute the dye in the upper water by using the action of the blower (wind) only. Do not mix in the dye with a spoon, as this action could disrupt the thermocline. If this step is done correctly, the upper warm waters will be stained blue and the lower waters will be clear. At this point, the stability of the thermocline can be explored by having the wind blow continuously across the surface. For further information on this demonstration, refer to the following sources:

- Vallentyne, J.R. 1967. A simplified model of a lake for instructional use. *J. Fish. Res. Bd. Canada* 24:2473–2479.
- Wetzel, R.G., and G.E. Likens. 2000. *Limnological Analysis*. 3d ed. New York: Springer-Verlag.

4. The concept called Q_{10} states that metabolic processes increase their rate by a factor of two for every rise in temperature of 10°C .
5. Differences in temperature and salinity combine to produce differences in water density, which are reflected in the separation of water masses on a grand scale into upper-water and deepwater masses. The upper-water mass is characterized by continuous motion. The use of wave tanks or a basin model might be very helpful to illustrate these concepts. Consult Wetzel and Likens (2000) for the details of such demonstrations.
6. The Coriolis effect (not a force) is often a puzzling concept for students to understand. Short of taking students to an amusement park to toss a ball back and forth on a merry-go-round, you might try setting up a large, flat, smooth plate on a lab stool that rotates. (Smooth wood will be adequate; Plexiglas™ might be better.) The plate should be marked with a grid. Marbles can be observed (in person or on videotape) as they are rolled across the surface. A shallow, grooved inclined plane will provide a uniform velocity for rolling the marble. (N.B.: I have found the commercial products advertised to demonstrate this phenomenon to be, in the main, overpriced and inadequate for the task.)
7. The Ekman spiral can be a difficult concept to illustrate in two dimensions. The figure in the textbook is more than adequate for most students, but a 3-D model made from wood or Plexiglas usually drives home the point for the remainder of the class.
8. In contrast to movements of the upper-water mass, movement of deep-water masses is ponderously slow. These waters are very cold and generally well oxygenated. Some further references on this topic include the following:

Broecker, W.S. 1994. An unstable superconveyor. *Nature* 367:414–416.

Broecker, W.S. 1995. Chaotic climate. *Sci. Am.* 273(5):62–68.

Lehman, S.J., and L.D. Keigwin. 1992. Deep-sea sediments reveal the history of the great ocean conveyor. *Oceanus* 35(4):89–92.

Needler, G.T. 1992. WOCE: The World Ocean Circulation Experiment. *Oceanus* 35(2):74–77.

III. Some Ecological Principles

The depth of coverage required for this part will depend on the prerequisites for the course. For students who have taken a general ecology course, this part will be a good review of some of the basic principles. It is broken down into five sections:

- Terms and Definitions
- Ecosystem Components
- Biogeochemical Cycles
- Biotic Structure of Ecosystems
- Ecological Control and Regulation

abiotic	ecology	herbivore
adsorption	ecosystem	heterotroph
autotroph	ecotone	inhibition model
biogeochemical cycle	energy flow	interference competition
biotic	energy pyramid	intermediate disturbance
<i>Calanus finmarchicus</i>	equilibrium model	interspecific competition
carnivore	exploitive competition	intraspecific competition
chitin	facilitation model	key industry species
climax community	food chain	keystone species
community	food web	krill
competition	fundamental niche	law of the minimum
competitive-exclusion principle	generalist	Leibig's law
copepod	grazer	limiting factor
decomposer	habitat	<i>Mytilus</i>
dominants	heat loss	niche

6 Instructor's Manual for Marine Biology

niche diversification	primary production	standing crop
nonequilibrium model	realized niche	succession
omnivore	reservoir	tolerance level
parasite	seres	tolerance model
periodic disturbance	specialist	trophic level
photosynthesis	species	trophic pyramid
phytoplankton	species diversity	trophic structure
population	species evenness	
predator	species richness	

1. Life maintains itself by capturing energy from a variety of sources (negative entropy). Nearly all of the world's ecosystems share a common energy source: the sun. Photosynthetic autotrophs capture light energy and use that energy to produce ATP and a reduced coenzyme, NADP (reduced). The biochemical reactions responsible for this process are the light reactions (also called light-dependent reactions). ATP and NADP (reduced) are then used in another set of reactions, called the dark reactions or the Calvin Benson cycle (also referred to as light-independent reactions). In that cycle, carbon dioxide is reduced (fixed) to a hexose sugar (glucose).
2. A few ecosystems have a different energy source than the sun. Thermal hot springs, whether terrestrial or aquatic, often release highly reduced compounds that may be oxidized (for example, H_2S). In this process, ATP and NADP (reduced) are produced. Bacteria capable of this type of metabolism are called nonphotosynthetic autotrophs.
3. Organisms capture energy, and, in a sense, they capture nutrients, too. Energy flows through ecosystems from trophic level to trophic level, but at each node, some energy becomes unavailable to the system; it is lost as heat. Energy flows through ecosystems, but nutrients cycle!
4. Species occupy recognizable regions and interact with the biotic and abiotic components of those places. Careful observation permits us to categorize geographic regions by the kind and number of species present and how they interact with one another through time.
5. Three important biogeochemical cycles (carbon, phosphorus, and nitrogen) are illustrated in the text. I have found that discussing the information presented in the biomass or energy pyramids in conjunction with one or more of the biogeochemical cycles can provide a better insight for the students. However, the nitrogen cycle always presents a challenge to students lacking an understanding of the oxidation states available to nitrogen (+5, +3, -3, and 0).
6. One subsection of this ecological primer explores the concepts of niche, species richness and diversity, and the limits of population growth. I have found that discussion of terrestrial examples helps students better understand these concepts. The same is true when reviewing the concepts of succession (that is, the facilitation, inhibition, and tolerance models).

Bertness, M.D., S.D. Gaines, and M.E. Hay. 2000. *Marine community ecology*. Sunderland, MA: Sinauer Associates, Inc., Publ.

IV. Larvae and Larval Ecology

Unless students have taken rather specialized courses, few will have been introduced to the concepts encompassed by the fascinating discipline of larval ecology. This part of Chapter 1 is divided into three sections as follows:

- Larval Types and Strategies
- Larval Ecology and Community Establishment
- Life History Strategies

brooders	iteroparous	lecithotrophic larvae
delayed metamorphosis	juveniles	migration
direct development larvae	K-selected species	nonpelagic larvae
equilibrium species	larval recruitment	opportunistic species

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|-----------------------|-------------|----------------------|
| pheromone | sedentary | substratum selection |
| planktotrophic larvae | semelparous | Thorson's model |
| r-selected species | sessile | Vance's model |
- Larval Types and Strategies*—Planktotrophic larvae possess little yolk and feed while swimming. Animals that produce this type of larva usually produce many eggs. On the other hand, some such animals produce fewer, energy (yolk)–rich eggs. Larvae that develop from these eggs (lecithotrophic) do not feed while swimming. Nonpelagic larvae hatch as fully formed miniatures of the adult and are called juveniles. Each of these developmental strategies has advantages and disadvantages. These issues are explored in depth in this section of the chapter.
 - Larval Ecology and Community Establishment*—Settlement behavior of planktonic larvae has evoked an enormous amount of research, and Nybakken reviews the classical works in the field. One of the pitfalls of studying these behaviors is the trap of falling into teleological speech or thinking, but forewarned students should be able to avoid that or at least explain that their wording is merely a matter of convenient speech. Both physical factors (for example, surface texture) and biotic factors (for example, pheromones) are important in larval settlement.
 - Life History Strategies*—Opportunistic species, called r-selected species, develop rapidly, have short life spans, and reproduce rapidly. Examples include single-celled algae, protozoans, copepods, and rotifers. Equilibrium species, called K-selected species, develop slowly, have longer life spans, and reproduce slowly. Examples include marine mammals.

V. Comparison of Terrestrial and Marine Ecosystems

This important part is divided into the following four sections:

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|---|--|--|
| ■ Physical and Chemical Differences | | |
| ■ Biodiversity Differences | | |
| ■ Life History Differences | | |
| ■ Structural and Functional Differences | | |
- | | | |
|------------------------|----------------|--------------------|
| cellulose | filter feeders | protein |
| community structure | floatation | sessile |
| copepods | food web links | skeletons |
| dissolved organic | gravity | sound |
| material (DOM) | krill | suspension feeders |
| electrical resistivity | plankton | wood |
- Physical and Chemical Differences*—Seawater is some 830× denser than air. This simple fact has an important consequence: There is no real counterpart of the plankton in terrestrial systems. Thus, filter feeders are unique to aquatic systems, except perhaps for spiders. Gravity has much less of an influence on aquatic organisms. This feature leads to profoundly different sorts of skeletal structures and energy storage materials. In general, terrestrial organisms expend considerable energy in the production of skeletons of carbohydrate and bone. This is generally not the case for aquatic organisms, which have protein as their dominate biochemical.
- Several other differences between aquatic and terrestrial systems are important to note. The speed of sound is 4× faster in water than in air, and the wavelengths are comparably longer, too. Water is 10¹⁶× more resistant to the transmission of electrical current than air, and with greater depth, water pressure increases much more rapidly than air pressure.
- Denny, M.W. 1990. Terrestrial versus aquatic biology: the medium and its message. *Amer. Zool.* 30:111–121.
- Water contains dissolved organic materials (DOMs) at concentrations great enough to be a significant energy source for some organisms. For example, corals can take up significant amounts of DOMs from their environment. Pizza shops and backyard BBQs notwithstanding, our atmosphere

8 Instructor's Manual for Marine Biology

does not contain sufficient amounts of DOMs to accomplish the same thing. Two classic references are listed as follows:

Sorokin, Y.I. 1973. On the feeding of some scleractinian corals with bacteria and dissolved organic matter. *Limnol. Oceanogr.* 18:380–385.

Stephens, G.C. 1962. Uptake of glucose from solution by the solitary coral *Fungia*. *Science* 131:1532–1534.

2. *Biodiversity Differences*—One way of comparing the biodiversity of terrestrial and marine systems is to consider the diversity of grazers or predators in the two systems. However, besides the diversity that one may recognize at higher taxonomic levels, there is also the consideration of just how much genetic diversity may be present in the oceans. This genetic diversity will probably be plumbed for its riches over the next several decades.
3. *Life History Differences*—Dispersal of gametes in the marine environment is very different than in terrestrial systems. For example, female gametes are never shed into the environment by terrestrial plants. Moreover, there are no marine counterparts to terrestrial pollinators.
4. *Structural and Functional Differences*—Large, long-lived flowering plants monopolize terrestrial ecosystems, whereas most of the marine environment is dominated by short-lived microscopic plants, which in turn are eaten by small grazing crustaceans (for example, protozoans, copepods, and krill). Terrestrial grazers often do not remove substantial amounts of the biomass, and their ability to digest terrestrial plants is limited. Digestion of such plants is accomplished only with the aid of an elaborate bacterial flora located in the gut. Marine grazers (for example, copepods, clams, and barnacles) consume the entire plant. Thus, terrestrial systems are dominated by conspicuous plant biomass, while in marine ecosystems the animal component is the most noticeable. This factor naturally leads to the use of descriptive terms that reflect the dominant members of the community—tall-grass prairies and coral reefs, for example.

VI. Divisions of the Marine Environment

“Comparisons are odorous.”

—Shakespeare, *Much Ado about Nothing*, 3:5:18

Nothing else will do; to carry out any intelligent discussion of the discipline, students must know these major subdivisions of the world ocean:

abyssal	disphotic	littoral
abyssalpelagic	epipelagic	mesopelagic
aphotic zone	epiphotic	neritic zone
bathyal	euphotic zone	oceanic zone
bathypelagic	hadal	pelagic
benthic	hadalpelagic	photic
benthos	intertidal zone	sublittoral

ENRICHMENT EXERCISES

Selected References

Following are some references that should be useful to get your students more involved in the concept of the marine environment. I have separated them into general references and works on ENSO.

General

Baretta-Bekker, J.G., E.K. Duursma, and R. Bouwe. 1992. *Encyclopedia of marine sciences*. New York: Springer-Verlag.

Castro, P., and M.E. Huber. 1992. *Marine biology*. St. Louis, MO: Mosby-Year Book, Inc.

- Cobb, J.S., and M.M. Harlin. 1976. *Marine ecology: selected readings*. Baltimore: University Park Press.
- Gebruk, A.V., E.C. Southward, and P.A. Tyler. 1997. *Advances in marine biology: the biogeography of the oceans*. San Diego: Academic Press.
- Jumas, P.A. 1993. *Concepts in biological oceanography: an interdisciplinary primer*. Oxford, U.K.: Oxford University Press.
- Kennish, M.J. 2000. *Practical handbook of marine science*. 3d ed. Boca Raton, FL: CRC Press.
- Kingsford, M., and C. Battershill. 2000. *Studying temperate marine environments*. Boca Raton, FL: CRC Press.
- Knox, G.A. 1994. *The biology of the Southern Ocean*. New York: Cambridge University Press.
- Kunzig, R. 1999. *The restless sea: Exploring the world beneath the waters*. New York: W.W. Norton.
- Lalli, C.M., and T.R. Parsons. 1993. *Biological oceanography: an introduction*. Oxford, U.K.: Pergamon Press Ltd.
- Nybakken, J.W. 1986. *Readings in marine ecology*. 2d ed. New York: Harper & Row.
- Ross, D.A. 1995. *Introduction to oceanography*. New York: HarperCollins, Publ.
- Stumm, W., and J.J. Morgan. 1996. *Aquatic chemistry: chemical, equilibria and rates in natural waters*. 3d ed. New York: Wiley.

El Niño and Other Water Movements

- Barber, R.T., and F.P. Chavez. 1983. Biological consequences of El Niño. *Science* 222:1203–1210.
- Canby, T.Y. 1984. El Niño's ill wind. *Nat. Geo.* 165(2):144–183.
- Cane, M.A. 1983. Oceanographic events during El Niño. *Science* 222:1189–1195.
- Graham, N.E., and W.B. White. 1988. The El Niño cycle: a natural oscillator of the Pacific Ocean–atmosphere system. *Science* 240:1293–1302.
- Glynn, P.W. 1988. El Niño's Southern Oscillation 1982–1983: nearshore population, community, and ecosystem responses. *Ann. Rev. Ecol. Syst.* 19:309–345.
- Glynn, P.W., and M.W. Colgan. 1992. Sporadic disturbances in fluctuating coral reef environments: El Niño and coral reef development in the Eastern Pacific. *Am. Zool.* 32:707–718.
- Longhurst, A. 1998. *Ecological geography of the sea*. San Diego: Academic Press.
- Philander, S.G. 1989. El Niño and La Niña. *Am. Sci.* 77:451–459.
- Philander, S.G. 1992. El Niño. *Oceanus* 35(2):56–61.
- Ramage, C.S. 1986. El Niño. *Sci. Am.* 254(6):77–83.
- Rasmusson, E.M., and J.M. Wallace. 1993. Meteorological aspects of the El Niño/Southern Oscillation. *Science* 222:1195–1202.
- Sandweiss, D.H., *et al.* 1996. Geoarchaeological Evidence from Peru for a 5000 Years B.P. Onset of El Niño. *Science* 273:1531–1533.
- Sellers, P.J., *et al.* 1997. Modeling the Exchanges of Energy, Water, and Carbon Between Continents and the Atmosphere. *Science* 275:502–509.
- van der Werf, G.R., *et al.* 2004. Continental-scale partitioning of fire emissions during the 1997 to 2001 El Niño/La Niña period. *Science* 303:73–76.

Great Oceanic Conveyor Belt

The following issue can be used as a discussion topic, perhaps as part of a cooperative learning exercise: Speculate as to what would happen if the Great Oceanic Conveyor Belt were to stop abruptly.

